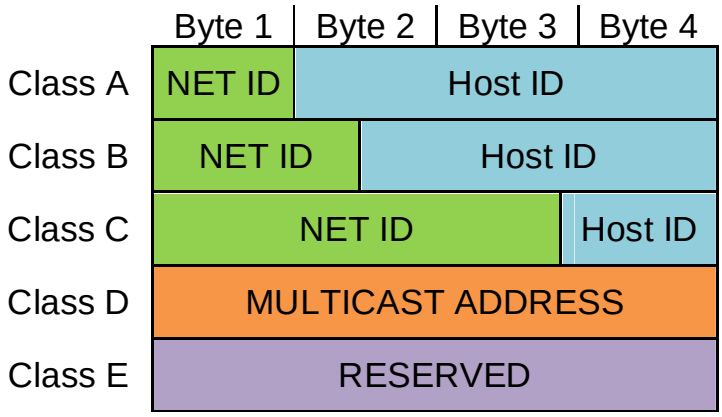
CORE – 9: Computer Networks (Unit–4)

Logical Addressing: Logical Addressing is one of the major responsibilities of the network layer. A logical address or IP address points to host / node / server or it can represent a whole network. These addresses can be changed by appropriate configurations. Logical addressing is needed for the purpose of routing between networks. A logical address can be IPv4 (32 bit) or IPv6 (128 bit).

IPV4 Addresses: An **Internet Protocol address** (**IP address**) is a numerical label assigned to each device connected to a computer network that uses the Internet Protocol for communication. These are usually represented in dot-decimal notation, consisting of four decimal numbers, each ranging from 0 to 255, separated by dots, e.g., 172.16.254.1.



Each part represents a group of 8 bits (an octet) of the address. An IPv4 address is divided into 5 sub-classes each class is divided into two parts:

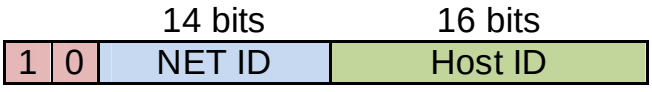
- **Network ID:** It represents the number of networks.
- **Host ID:** It represents the number of hosts.

Class A: In Class A, The network ID is 8 bits long and the host ID is 24 bits long.



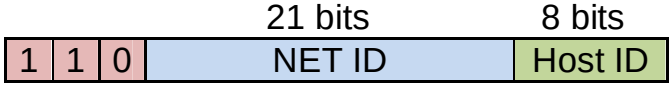
The first bit in higher order bits of the first octet is always set to 0 and the remaining 7 bits determine the network ID. The 24 bits determine the host ID in any network. The total number of networks in Class A is $2^7 - 2 = 126$ network address (all 0s and all 1s are not taken) i.e. the network addresses ranges from **1.x.x.x to 126.x.x.x** only. The IP range **127.x.x.x is reserved for loopback IP addresses**. The default subnet mask for Class A is 255.0.0.0. The total number of hosts in each Class A network is **$2^{24} - 2$** (all 0s and all 1s are not taken).

Class B: In Class B, The Network ID is 16 bits long and The Host ID is 16 bits long.



The higher order bits of the first octet are always set to 10, and the remaining14 bits determine the network ID. The other 16 bits determine the Host ID.
The total number of networks in Class B is 2^{14} i.e. the the network addresses range from 128.0.x.x to 191.255.x.x. The default subnet mask for Class B is 255.255.x.x.
The total number of hosts in each Class B is $2^{16} - 2$ (all 0s and all 1s are not taken).

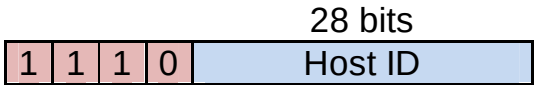
Class C: In Class C, the Network ID is 24 bits long and the host ID is 8 bits long.



The higher order bits of the first octet are always set to 110, and the remaining 21 bits determine the network ID. The 8 bits of the host ID determine the host in a

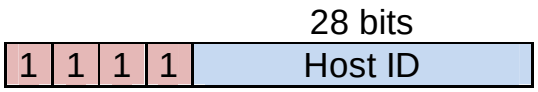
network.
The higher order bits of the first octet are always set to 110, and the remaining 21 bits determine the network ID. The 8 bits of the host ID determine the host in a network.
The total number of networks in Class C is 2^{21} i.e. the network addresses range from 192.0.0.x to 223.255.255.x. The default subnet mask for Class C is 255.255.255.x.
The total number of hosts each Class C network is 2^8 - 2 = 254 host address

Class D: In Class D, an IP address is reserved for multicast addresses. It does not possess subnetting.



The higher order bits of the first octet are always set to 1110, and the remaining bits determine the host ID in any network.

Class E: In Class E, an IP address is used for the future use or for the research and development purposes. It does not possess any subnetting.



The higher order bits of the first octet are always set to 1111 and the remaining bits determine the host ID in any network.

Rules for assigning Host ID: The Host ID is used to determine the host within any network. The Host ID is assigned based on the following rules:

- The Host ID must be unique within any network.
- The Host ID in which all the bits are set to 0 cannot be assigned as it is used to represent the network ID of the IP address.
- The Host ID in which all the bits are set to 1 cannot be assigned as it is reserved for the multicast address.

Classful Network Architecture:

Class	Higher bits	No. of NET ID bits	No. of HOST ID bits	No. of networks	No.of hosts per network	Range
A	0	8	24	2^7 - 2	2^{24} - 2	0.0.0.0 to 127.255.255.255
B	10	16	16	2^{14}	2^{16} - 2	128.0.0.0 to 191.255.255.255
C	110	24	8	2^{21}	2^8 - 2	192.0.0.0 to 223.255.255.255
D	1110	Not Defined	Not Defined	Not Defined	Not Defined	224.0.0.0 to 239.255.255.255
E	1111	Not Defined	Not Defined	Not Defined	Not Defined	240.0.0.0 to 255.255.255.255

IPV6 Addresses: An IPv6 address is made of 128 bits divided into eight bolocks of 16-bits each. Each block is then converted into 4-digit Hexadecimal numbers separated by colon symbols.

Example - 2001: 0000: 3238: DFE1: 0063: 0000: 0000: FEFB

IPv6 provides some rules to shorten the address. The rules are as follows:

Rule.1: All Leading zeros are discarded.

Rule.2: if a block contain all zeros it shrunks down to a single zero.

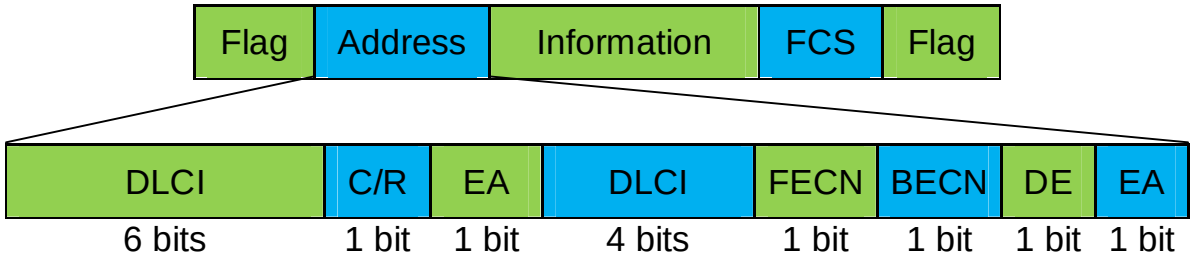
Rule.3: If two of more blocks contain consecutive zeroes, they are omitted and are replaced with double colon sign ::

Ex - 2001: 0000: 3238: DFE1: 0063: 0000: 0000: FEFB is written as 2001:0:3238:DFE1:63:: FEFB

Virtual-Circuit Networks: A virtual circuit (VC) is a means of transporting data over a packet-switched network in such a way that it appears as if there is a dedicated physical link between the source and destination. The term virtual circuit is synonymous with virtual connection.

Frame Relay: Frame Relay is a standardized wide area network technology that specifies the physical and data link layers of digital telecommunications channels using a packet switching methodology.

Frame Rlay Frame Format:



DLCI: Data Link Connection Identifier
EA: Extended Address
DE: Discard Eligibility
C/R: Command/Response
FECN: Forword Explicit Congession Notification
BECN: Backword Explicit Congession Notification

Each Frame Relay protocol data unit (PDU) consists of the following fields:

1. **Flag Field**. The flag is used to perform high-level data link synchronization which indicates the beginning and end of the frame with the unique pattern 01111110.
2. **Address Field**. Each address field may occupy octet 2 to 3, octet 2 to 4, or octet 2 to 5, depending on the range of the address in use. A two-octet address field is shown in the above figure.
    i. **DLCI** (Data Link Connection Identifier Bits) - The DLCI serves to identify the virtual connection so that the receiving end knows which information connection a frame belongs to. Note that this DLCI has only local

significance. A single physical channel can multiplex several different virtual connections.

ii. **C/R** field indicates whether the frame is acommand frame or response frame.

iii. **EA** field denotes the address field extension bit.

iv. **FECN, BECN, DE** bits- These bits report congestion.

- **FECN** = Forward Explicit Congestion Notification bit
- **BECN** = Backward Explicit Congestion Notification bit
- **DE** = Discard Eligibility bit

3. **Information Field**. A system parameter defines the maximum number of data bytes that a host can pack into a frame. Hosts may negotiate the actual maximum frame length at call set-up time. The standard specifies the maximum information field size (supportable by any network) as at least 262 octets. Since end-to-end protocols typically operate on the basis of larger information units, Frame Relay recommends that the network support the maximum value of at least 1600 octets in order to avoid the need for segmentation and reassembling by end-users.
4. **Frame Check Sequence (FCS) Field**. Since one cannot completely ignore the bit error-rate of the medium, each switching node needs to implement error detection to avoid wasting bandwidth due to the transmission of *err*ed frames. The error detection mechanism used in Frame Relay uses the cyclic redundancy check (CRC) as its basis.

ATM: ATM stands for Asynchronous Transfer Mode. It is a switching technique that uses time division multiplexing (TDM) for data communications. ATM networks are connection oriented networks for cell relay that supports voice, video and data communications. It encodes data into small fixed - size cells so that they are suitable for TDM and transmits them over a physical medium. The size of an ATM cell is 53 bytes: 5 byte header and 48 byte payload.

Benefits of ATM Networks:

- It provides the dynamic bandwidth that is particularly suited for bursty traffic.
- Since all data are encoded into identical cells, data transmission is simple, uniform and predictable.
- Uniform packet size ensures that mixed traffic is handled efficiently.
- Small sized header reduces packet overload, thus ensuring effective bandwidth usage.
- ATM networks are scalable both in size and speed.

Transport Layer: Various responsibilities of a Transport Layer are –

- Process to process delivery
- End-to-end Connection between hosts
- Multiplexing and Demultiplexing
- Congestion Control
- Flow Control
- Data integrity and Error correction

- **Process to process delivery –** While Data Link Layer requires the MAC address (48 bits address contained inside the Network Interface Card of every host machine) of source-destination hosts to correctly deliver a frame and Network layer requires the IP address for appropriate routing of packets, in a similar way Transport Layer requires a **Port number** to correctly deliver the segments of data to the correct process amongst the multiple processes running on a particular host. A **port number** is a 16 bit address used to identify any client-server program uniquely.
- **End-to-end Connection between hosts –** The transport layer is also responsible for creating the end-to-end Connection between hosts for which it mainly uses TCP and UDP. TCP is a secure, connection- orientated protocol which uses a handshake protocol to establish a robust connection between two end- hosts. TCP ensures reliable delivery of messages and is used in various applications. UDP, on the other hand, is a stateless and unreliable protocol which ensures best-effort delivery. It is suitable for the applications which have little concern with flow or error control and requires sending the bulk of data like video conferencing. It is often used in multicasting protocols.
- **Multiplexing and Demultiplexing –** Multiplexing allows simultaneous use of different applications over a network which is running on a host. The transport layer provides this mechanism which enables us to send packet streams from various applications simultaneously over a network. The transport layer accepts these packets from different processes differentiated by their port numbers and passes them to the network layer after adding proper headers. Similarly, Demultiplexing is required at the receiver side to obtain the data coming from various processes. Transport layer receives the segments of data from the network layer and delivers it to the appropriate process running on the receiver's machine.
- **Congestion Control –** Congestion is a situation in which too many sources over a network attempt to send data and the router buffers start overflowing due to which loss of packets occur. As a result retransmission of packets from the sources increases the congestion further. In this situation, the Transport layer provides Congestion Control in different ways. It uses **open loop** congestion control to prevent the congestion and **closed loop** congestion control to remove the congestion in a network once it occurred. **TCP provides AIMD- additive increase multiplicative decrease**, **leaky bucket** technique for congestion control.
- **Flow control –** The transport layer provides a flow control mechanism between the adjacent layers of the TCP/IP model. TCP also prevents data loss due to a fast sender and slow receiver by imposing some flow control techniques. It uses the method of sliding window protocol which is accomplished by the receiver by sending a window back to the sender informing the size of data it can receive.
- **Data integrity and Error correction –** By means of an error detection code, such as a checksum, the transport protocol may check that the data is not corrupted, and verify correct receipt by sending an ACK or NACK message to

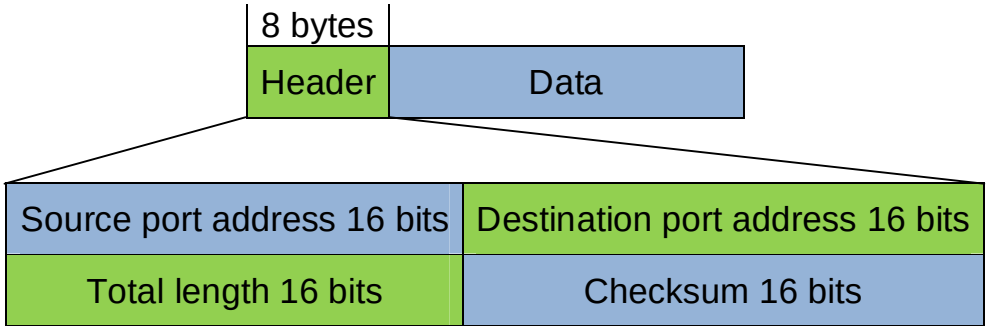
the sender. Automatic repeat request schemes may be used to retransmit lost or corrupted data.

Transport Layer protocols: The two protocols of transport layer are UDP and TCP.

UDP

- UDP stands for **User Datagram Protocol**.
- UDP is a connectionless protocol.
- This type of protocol is used when reliability and security are less important than speed and size.
- UDP is an end-to-end transport level protocol that adds transport-level addresses, checksum error control, and length information to the data from the upper layer.
- The packet produced by the UDP protocol is known as a user datagram.

UDP Frame Format: The UDP frame format is shown in the figure below.



Where,

- **Source port address:** It defines the address of the application process that has delivered a message. The source port address is of 16 bits address.
- **Destination port address:** It defines the address of the application process that will receive the message. The destination port address is of a 16-bit address.
- **Total length:** It defines the total length of the user datagram in bytes. It is a 16-bit field.
- **Checksum:** The checksum is a 16-bit field which is used for error detection.

Disadvantages of UDP protocol

- It does not provide any sequencing or reordering functions and does not specify the damaged packet when reporting an error.
- UDP can discover that an error has occurred, but it does not specify which packet has been lost as it does not contain an ID or sequencing number of a particular data segment.
- The UDP message can be lost, delayed, duplicated, or can be out of order.
- It does not provide a reliable transport delivery service. It does not provide any acknowledgment or flow control mechanism. However, it does provide error control to some extent.

TCP

- TCP stands for Transmission Control Protocol.
- It is a connection-oriented protocol means the connection is established between both the ends of the transmission. For creating the connection, TCP generates a virtual circuit between sender and receiver for the duration of a transmission.

Features of TCP protocol:

- **Stream data transfer:** TCP protocol transfers the data in the form of contiguous stream of bytes. TCP group the bytes in the form of TCP segments and then passed it to the IP layer for transmission to the destination. TCP itself segments the data and forward to the IP.
- **Reliability:** TCP assigns a sequence number to each byte transmitted and expects a positive acknowledgement from the receiving TCP. If ACK is not received within a timeout interval, then the data is retransmitted to the destination. The receiving TCP uses the sequence number to reassemble the segments if they arrive out of order or to eliminate the duplicate segments.
- **Flow Control:** When receiving TCP sends an acknowledgement back to the sender indicating the number the bytes it can receive without overflowing its internal buffer. The number of bytes is sent in ACK in the form of the highest sequence number that it can receive without any problem. This mechanism is also referred to as a window mechanism.
- **Multiplexing:** Multiplexing is a process of accepting the data from different applications and forwarding to the different applications on different computers. At the receiving end, the data is forwarded to the correct application. This process is known as demultiplexing. TCP transmits the packet to the correct application by using the logical channels known as **ports**.
- **Logical Connections:** The combination of sockets (IP address + Port number), sequence numbers, and window sizes, is called a logical connection. Each connection is identified by the pair of sockets used by sending and receiving processes.
- **Full Duplex:** TCP provides Full Duplex service, i.e., the data flow in both the directions at the same time. To achieve Full Duplex service, each TCP should have sending and receiving buffers so that the segments can flow in both the directions. TCP is a connection-oriented protocol. Suppose the process A wants to send and receive the data from process B. The following steps occur:
    - Establish a connection between two TCPs.
    - Data is exchanged in both the directions.
    - The Connection is terminated.

TCP Segment Format:

<table>
<tr><td colspan="8">Source port address 16 bits</td><td colspan="8">Destination port address 16 bits</td></tr>
<tr><td colspan="16">Sequence number 32 bits</td></tr>
<tr><td colspan="16">Acknowledgement 32 bits</td></tr>
<tr><td colspan="2">HLEN
4 bits</td><td colspan="3">Reserved
6 bits</td><td>URG</td><td>ACK</td><td>PSH</td><td>RST</td><td>SYN</td><td>FIN</td><td colspan="8">Window size 16 bits</td></tr>
<tr><td colspan="8">Checksum 16 bits</td><td colspan="8">Urgent pointer 16 bits</td></tr>
<tr><td colspan="16">Option and padding up to 40 bytes</td></tr>
</table>

The header of a TCP segment can range from 20-60 bytes. 40 bytes are for options. If there are no options, header is of 20 bytes else it can be maximum of 60 bytes.

Header fields:

- **Source Port Address –** 16 bit field that holds the port address of the application that is sending the data segment.
- **Destination Port Address –** 16 bit field that holds the port address of the application in the host that is receiving the data segment.
- **Sequence Number –** 32 bit field that holds the sequence number, i.e. the byte number of the first byte that is sent in that particular segment. It is used to reassemble the message at the receiving end if the segments are received out of order.
- **Acknowledgement Number –** 32 bit field that holds the acknowledgement number, i.e, the byte number that the receiver expects to receive next. It is an acknowledgment for the previous bytes being received successfully.
- **Header Length (HLEN) –** This is a 4 bit field that indicates the length of the TCP header by number of 4-byte words in the header, i.e, if the header is of 20 bytes (min length of TCP header), then this field will hold 5 (because 5 x 4 = 20) and the maximum length: 60 bytes, then it'll hold the value 15(because 15 x 4 = 60). Hence, the value of this field is always between 5 and 15.
- **Control flags –** These are 6 1-bit control bits that control connection establishment, connection termination, connection abortion, flow control, mode of transfer etc. These are -
  - **URG:** The URG field indicates that the data in a segment is urgent.
  - **ACK:** When ACK field is set, then it validates the acknowledgement number.
  - **PSH:** The PSH field is used to inform the sender that higher throughput is needed so if possible, data must be pushed with higher throughput.
  - **RST:** The reset bit is used to reset the TCP connection when there is any confusion occurs in the sequence numbers.
  - **SYN:** The SYN field is used to synchronize the sequence .

- **FIN:** The FIN field is used to inform the receiving TCP module that the sender has finished sending data. It is used in **connection termination** in three types of segments: termination request, termination confirmation, and acknowledgement of termination confirmation.

- **Window size –** This field tells the window size of the sending TCP in bytes.
- **Checksum –** This field holds the checksum for error control. It is mandatory in TCP as opposed to UDP.
- **Urgent pointer –** This field (valid only if the URG control flag is set) is used to point to data that is urgently required that needs to reach the receiving process at the earliest. The value of this field is added to the sequence number to get the byte number of the last urgent byte.

Differences between TCP & UDP

TCP	UDP
TCP establishes a virtual circuit before transmitting the data.	UDP transmits the data directly to the destination computer without verifying whether the receiver is ready to receive or not.
It is a Connection-Oriented protocol	It is a Connectionless protocol
It is a reliable protocol.	It is an unreliable protocol.
Header size is 20 bytes	Header size is 8 bytes
It waits for the acknowledgement of data and has the ability to resend the lost packets.	It neither takes the acknowledgement nor does it retransmit the damaged frame.

Application Layer: Various protocols in application layer are DNS, SMTP, POP, FTP, HTTP etc.

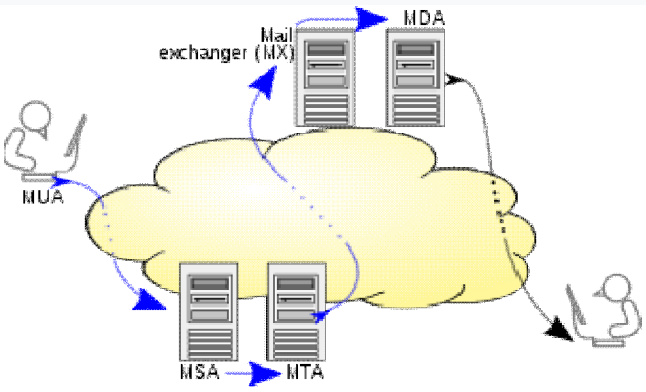
DNS: DNS stands for Domain Name System. It is a directory service that provides a mapping between the name of a host on the network and its numerical address. Each node in a network has a domain name, and a full domain name is a sequence of symbols specified by dots. DNS is a service that translates the domain name into IP addresses. This allows the users of networks to utilize user-friendly names when looking for other hosts instead of remembering the IP addresses. For example, suppose the site at iitkgp.ac.in had an IP address of 132.147.165.50, most people would reach this site by specifying iitkgp.ac.in. Therefore, the domain name is more reliable than IP address.

Working of DNS

- DNS is a client/server network communication protocol. DNS clients send requests to the server while DNS servers send responses to the client.
- Client requests contain a name which is converted into an IP address known as a forward DNS lookups while requests containing an IP address which is converted into a name known as reverse DNS lookups.
- DNS implements a distributed database to store the name of all the hosts available on the internet.
- If a client like a web browser sends a request containing a hostname, then a piece of software such as **DNS resolver** sends a request to the DNS server to obtain the IP address of a hostname. If DNS server does not contain the IP address associated with a hostname, then it forwards the request to another DNS server. If IP address has arrived at the resolver, which in turn completes the request over the internet protocol.

SMTP: **The** Simple Mail Transfer Protocol **(**SMTP**)** is a communication protocol for electronic mail transmission.

Mail processing model: Email is submitted by a mail client (mail user agent, **MUA**) to a mail server (mail submission agent, **MSA**) using SMTP on TCP port 587. Most mailbox providers still allow submission on traditional port 25. The MSA delivers the mail to its mail transfer agent (mail transfer agent, **MTA**). Often, these two agents are instances of the same software launched with different options on the same machine. Local processing can be done either on a single machine, or split among multiple machines; mail agent processes on one machine can share files, but if processing is on multiple machines, they transfer messages between each other using SMTP, where each machine is configured to use the next machine as a smart host. Each process is an MTA (an SMTP server) in its own right.



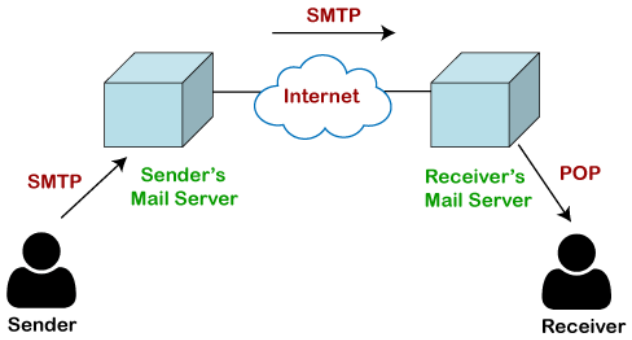
The boundary MTA uses DNS to look up the MX (**mail exchanger**) record for the recipient's domain (the part of the email address on the right of @). The MX record contains the name of the target MTA. Based on the target host and other factors, the sending MTA selects a recipient server and connects to it to complete the mail exchange.

Message transfer can occur in a single connection between two MTAs, or in a series of hops through intermediary systems. A receiving SMTP server may be the ultimate destination, an intermediate "relay" (that is, it stores and forwards the message) or a "gateway" (that is, it may forward the message using some protocol other than SMTP). Each hop is a formal handoff of responsibility for the message, whereby the receiving server must either deliver the message or properly report the failure to do so.

Once the final hop accepts the incoming message, it hands it to a mail delivery agent (**MDA**) for local delivery. An MDA saves messages in the relevant mailbox format. As with sending, this reception can be done using one or multiple computers, but in the above diagram, the MDA is depicted as one box near the mail exchanger box. An MDA may deliver messages directly to storage, or forward them over a network using SMTP or other protocol such as Local Mail Transfer Protocol (LMTP) for this purpose.

Once delivered to the local mail server, the mail is stored for batch retrieval by authenticated mail clients (MUAs). Mail is retrieved by end-user applications, called email clients, using Internet Message Access Protocol (IMAP), a protocol that both facilitates access to mail and manages stored mail, or the Post Office Protocol (**POP**) which typically uses the traditional mbox mail file format or a proprietary system such as Microsoft Exchange/Outlook or Lotus Notes/Domino. Webmail clients may use either method, but the retrieval protocol is often not a formal standard.

POP: POP stands for Post Office Protocol. It is an application-layer Internet standard protocol used by e-mail clients to retrieve e-mail from a mail server.



- POP is an application layer internet standard protocol.
- Since POP supports offline access to the messages, thus requires less internet usage time.
- POP does not allow search facility.
- In order to access the messaged, it is necessary to download them.
- It allows only one mailbox to be created on server.
- It is not suitable for accessing non mail data.
- POP commands are generally abbreviated into codes of three or four letters. Eg. STAT.

FTP:

- FTP stands for File transfer protocol.
- FTP is a standard internet protocol provided by TCP/IP used for transmitting files from one host to another.
- It is mainly used for transferring the web page files from their creator to the computer that acts as a server for other computers on the internet.
- It is also used for downloading the files to computer from other servers.

Why FTP? Although transferring files from one system to another is very simple and straightforward, but sometimes it can cause problems. For example, two systems may have different file conventions. Two systems may have different ways to represent text and data. Two systems may have different directory structures. FTP protocol overcomes these problems by establishing two connections between hosts. One connection is used for data transfer, and another connection is used for the control connection.

Types of connections in FTP:

- **Control Connection:** The control connection uses very simple rules for communication. Through control connection, we can transfer a line of command or line of response at a time. The control connection is made between the control processes. The control connection remains connected during the entire interactive FTP session.
- **Data Connection:** The Data Connection uses very complex rules as data types may vary. The data connection is made between data transfer processes. The data connection opens when a command comes for transferring the files and closes when the file is transferred.

Advantages of FTP:

- **Speed:** One of the biggest advantages of FTP is speed. The FTP is one of the fastest ways to transfer the files from one computer to another computer.
- **Efficient:** It is more efficient as we do not need to complete all the operations to get the entire file.
- **Security:** To access the FTP server, we need to login with the username and password. Therefore, we can say that FTP is more secure.

Disadvantages of FTP:

- FTP serves two operations, i.e., to send and receive large files on a network. However, the size limit of the file is 2GB that can be sent. It also doesn't allow you to run simultaneous transfers to multiple receivers.
- Passwords and file contents are sent in clear text. So, it is quite possible that attackers can carry out the brute force attack by trying to guess the FTP password.
- It is not compatible with every system.

HTTP

- HTTP stands for **HyperText Transfer Protocol**.
- It is a protocol used to access the data on the World Wide Web (www).
- The HTTP protocol can be used to transfer the data in the form of plain text, hypertext, audio, video, and so on.
- This protocol is known as HyperText Transfer Protocol because of its efficiency that allows us to use in a hypertext environment where there are rapid jumps from one document to another document.
- HTTP is similar to the FTP as it also transfers the files from one host to another host. But, HTTP is simpler than FTP as HTTP uses only one connection, i.e., no control connection to transfer the files.

- HTTP is similar to SMTP as the data is transferred between client and server. The HTTP differs from the SMTP in the way the messages are sent from the client to the server and from server to the client. SMTP messages are stored and forwarded while HTTP messages are delivered immediately.
- It is a connectionless protocol.
- HTTP client initiates a request and waits for a response from the server. When the server receives the request, the server processes the request and sends back the response to the HTTP client after which the client disconnects the connection. The connection between client and server exist only during the current request and response time only.

Network Security: Network security consists of the policies and practices adopted to prevent and monitor unauthorized access, misuse, modification, or denial of a computer network and network-accessible resources.

Aspects of Network Security:

- **Privacy:** Privacy means both the sender and the receiver expects confidentiality. The transmitted message should be sent only to the intended receiver. Therefore, there is a requirement to encrypt the message so that the message cannot be intercepted.
- **Message Integrity:** Data integrity means that the data must arrive at the receiver exactly as it was sent. There must be no changes in the data content during transmission, either maliciously or accident, in a transit.
- **Authentication:** Authentication means that the receiver is sure of the sender's identity, i.e., no imposter has sent the message.
- **Non-Repudiation:** Non-Repudiation means that the receiver must be able to prove that the received message has come from a specific sender. The sender must not deny sending a message that he or she send.

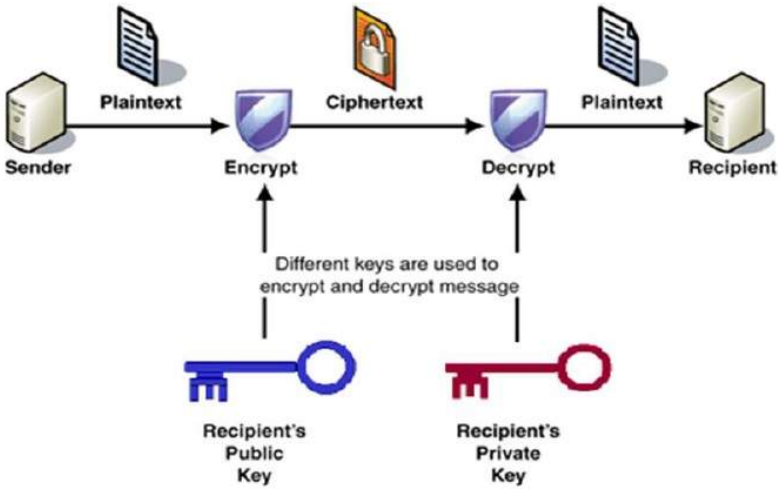
Encryption: Encryption means that the sender converts the original information into another form and sends the unintelligible message over the network.

Decryption: Decryption reverses the Encryption process in order to transform the message back to the original form.

The data which is to be encrypted at the sender site is known as **plaintext**, and the encrypted data is known as **ciphertext**. The data is decrypted at the receiver site.

Public Key Cryptography: The most important properties of public key encryption scheme are −

- Different keys are used for encryption and decryption.
- Each receiver possesses a unique decryption key, generally referred to as his private key.

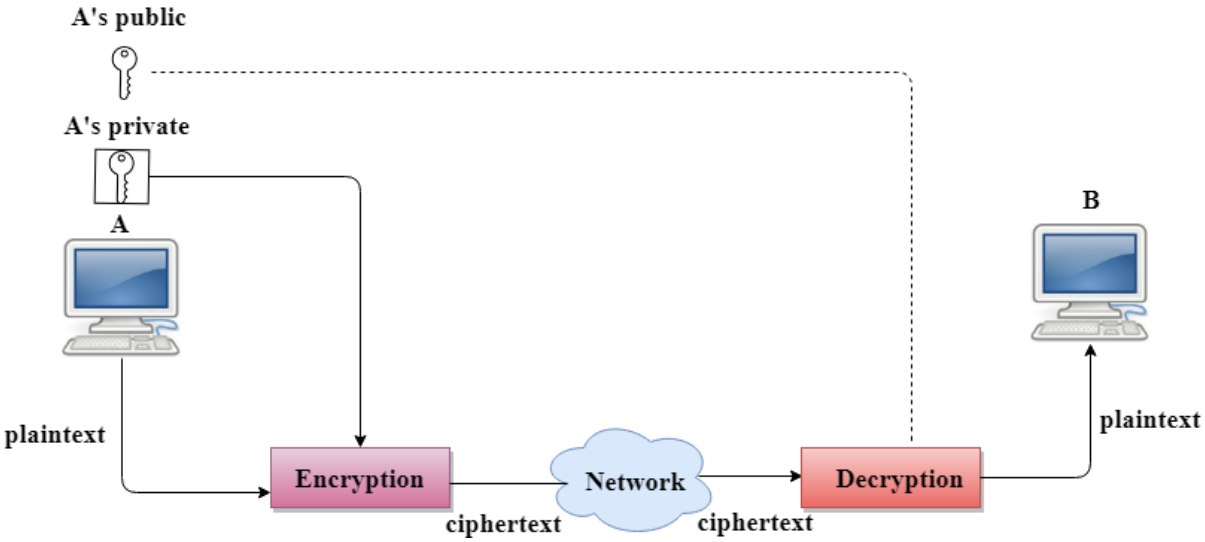


- Receiver needs to publish an encryption key, referred to as his public key.
- Some assurance of the authenticity of a public key is needed in this scheme to avoid spoofing by adversary as the receiver. Generally, this type of cryptosystem involves trusted third party which certifies that a particular public key belongs to a specific person or entity only.
- Encryption algorithm is complex enough to prohibit attacker from deducing the plaintext from the ciphertext and the encryption (public) key.
- Though private and public keys are related mathematically, it is not be feasible to calculate the private key from the public key. In fact, intelligent part of any public-key cryptosystem is in designing a relationship between two keys.

Digital Signature: The Digital Signature is a technique which is used to validate the authenticity and integrity of the message. We know that there are four aspects of security: privacy, authentication, integrity, and non-repudiation. Privacy is achieved by encryption and decryption. The other three can be achieved by using a digital signature. The basic idea behind the Digital Signature is to sign a document. When we send a document electronically, we can also sign it. We can sign a document in two ways: to sign a whole document and to sign a digest.

Signing the Whole Document

- In Digital Signature, a public key encryption technique is used to sign a document. However, the roles of a public key and private key are different here. The sender uses a private key to encrypt the message while the receiver uses the public key of the sender to decrypt the message.
- In Digital Signature, the private key is used for encryption while the public key is used for decryption.
- Digital Signature cannot be achieved by using secret key encryption.



Digital Signature is used to achieve the following three aspects:

- **Integrity:** The Digital Signature preserves the integrity of a message because, if any malicious attack intercepts a message and partially or totally changes it, then the decrypted message would be impossible.
- **Authentication:** We can use the following reasoning to show how the message is authenticated. If an intruder (user X) sends a message pretending that it is coming from someone else (user A), user X uses her own private key to encrypt the message. The message is decrypted by using the public key of user A. Therefore this makes the message unreadable. Encryption with X's private key and decryption with A's public key results in garbage value.
- **Non-Repudiation:** Digital Signature also provides non-repudiation. If the sender denies sending the message, then her private key corresponding to her public key is tested on the plaintext. If the decrypted message is the same as the original message, then we know that the sender has sent the message.

Signing the Digest

- Public key encryption is efficient if the message is short. If the message is long, a public key encryption is inefficient to use. The solution to this problem is to let the sender sign a digest of the document instead of the whole document.
- The sender creates a miniature version (digest) of the document and then signs it, the receiver checks the signature of the miniature version.
- The hash function is used to create a digest of the message. The hash function creates a fixed-size digest from the variable-length message.
- The two most common hash functions used are - MD5 (Message Digest 5) and SHA-1 (Secure Hash Algorithm 1).
- A hash function must have two properties to ensure the success:
    - First, the digest must be one way, i.e., the digest can only be created from the message but not vice versa.
    - Second, hashing is a one-to-one function, i.e., two messages should not create the same digest.

Following are the steps taken to ensure security:

- The miniature version (digest) of the message is created by using a hash function.
- The digest is encrypted by using the sender's private key.
- After the digest is encrypted, then the encrypted digest is attached to the original message and sent to the receiver.
- The receiver receives the original message and encrypted digest and separates the two. The receiver implements the hash function on the original message to create the second digest, and it also decrypts the received digest by using the public key of the sender. If both the digests are same, then all the aspects of security are preserved.

